



UNPOISON
EARTH • AIR • WATER

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28 April 2026

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RE: Submission of comments on biopesticides regulatory gaps in South Africa in relation to the Fertilizers, Farm Feeds, Agricultural Remedies and Stock Remedies Act, No. 36 of 1947.

1. These submissions are made on behalf of UnPoison, a multi-sector civil society movement with the goal of effecting legislative reform of the agricultural remedies framework within South Africa.
2. The submissions intend to highlight and advocate for the recognition of biopesticides as a separate regulatory category and guide the development of a targeted registration process separate from the conventional chemical pesticides largely inline with SADC guidelines and best practice from around the world.
3. This submission includes the following sections:
 - 3.1. Introduction and problem statement;
 - 3.2. Current regulatory framework within South Africa;
 - 3.3. International comparison of biopesticides/biological regulatory frameworks;
 - 3.4. Best practices for registration of biopesticides through comparison of foreign legislation;
 - 3.5. **Key elements and recommendations;**
 - 3.6. Conclusion

I Introduction and Problem Statement

4. To safeguard against pest infestations, huge crop losses, poor product quality and unsellable/unmarketable products, farmers will often apply chemical pesticides. Unfortunately, use of chemical pesticides exclusively, has led to adverse effects to human, animal and environmental health. The South African agricultural industry encompasses a broad spectrum of farming activities and contributes significantly to the country's GDP, employment and food security.¹ Its importance, influence and impact in South Africa, particularly the people and the environment cannot be understated. Through section 24² our Constitution recognises that “everyone has the right-

4.1. to an environment that is not harmful to their health and wellbeing; and

4.2. to have the environment protected for the benefit of present and future generations through reasonable legislative and other measures that-

4.2.1. prevent pollution and ecological degradation;

4.2.2. promote conservation; and

4.2.3. secure ecologically sustainable development and use of natural resources while promoting justifiable economic and social development.”

An Integrated Pest Management (IPM) strategy that encompasses various readily accessible control methods is widely encouraged.

5. In accordance with the Harmonised Guidelines for the Registration of Biopesticide Products and Biological Control Agents in Southern Africa, biopesticides is a generic term applied to substances derived from nature and applied in a manner similar to a conventional chemical pesticides that is normally used for short-term pest control.
6. In the South African context, biopesticides would fall under the term “biological remedy” included in the definition of an agricultural remedy. South Africa's current legal framework does not align to international best practice, is not representative of public interest, and is in dire need of reform.

¹ [Agricultural Overview and Stats in South Africa](#) posted by FarmingSA on 11 September 2023, accessed on 28 October 2025

² Constitution of the Republic of South Africa, 1996, section 24

7. An outdated, complex and unclear legal framework alongside a lack of focused strategy has created confusion around data requirements, scientific evaluations, decision-making and post-registration processes when applying for the registration of a biopesticide within South Africa.

II Current regulatory framework within South Africa

8. The Fertilizers, Farm Feeds, Agricultural Remedies and Stock Remedies Act, No 36 of 1947 (the “Act”) aims to regulate agricultural remedies that control agricultural pests. Such agricultural remedies can be harmful to human health, animals and the environment.

9. Per section 1 of the Act, an agricultural remedy is defined as:

“... any chemical substance or biological remedy, or any mixture or combination of any substance or remedy intended or offered to be used—

(a) for the destruction, control, repelling, attraction or prevention of any undesired microbe, alga, nematode, fungus, insect, plant, vertebrate, invertebrate, or any product thereof, but excluding any chemical substance, biological remedy or other remedy in so far as it is controlled under the Medicines and Related Substances Control Act, 1965 (Act No. 101 of 1965), or the Hazardous Substances Act, 1973 (Act No. 15 of 1973); or

(b) as plant growth regulator, defoliant, desiccant or legume inoculant, and anything else which the Minister has by notice in the Gazette declared an agricultural remedy for the purposes of this Act;”

10. Section 3 of the Act³ provides the framework for the registration process requiring that the Registrar be satisfied that the agricultural remedy applied for is:

- 10.1 suitable and sufficiently effective for the purposes for which it was intended, per the application;
- 10.2 it complies with prescribed requirements and;
- 0.3 it is not contrary to the public interest⁴.

11. The Act requires that the above must be met for all renewal registrations⁵.

12. Section 23 of the Act empowers the Minister to make regulations that prescribe the manner in which agricultural remedies are registered, the manner in which thereof is renewed and the information the applicant is required to furnish alongside the application, whether new or renewal.

13. The Act, however, fails to set out explicit decision-making and administrative timeframes for biological remedies. The absence of specified deadlines, uncertainty of the duration of the registration and lack of a clear process for the registration of biological remedies has discouraged the biopesticide industry and product development.

14. The Regulations Relating to Agricultural Remedies, 2024 (the Regulations) sets out in detail the process to be followed when applying for the registration of an agricultural remedy:

³ “3(1)(a) Application for registration of a fertilizer, farm feed, agricultural remedy, stock remedy, sterilizing plant or pest control operator shall be made to the registrar in the prescribed manner and shall be accompanied by the prescribed application fee. (2) If, after consideration of any such application and after such investigation and enquiry as he may deem necessary, the registrar is satisfied that—

(a) the fertilizer, farm feed, agricultural remedy or stock remedy in respect of which registration is applied for is suitable and sufficiently effective for the purposes for which it is intended, and complies with such requirements as may be prescribed, and that it is not contrary to the public interest that it be registered, and that the establishment where it is manufactured is suitable for such manufacture, he shall register such fertilizer, farm feed, agricultural remedy or stock remedy;

(b) the sterilizing plant in respect of which registration is applied for is suitable and sufficiently effective for the purpose for which it is intended, and complies with such requirements as may be prescribed, and that it is not contrary to the public interest that such sterilizing plant be registered, he shall register such sterilizing plant;”

⁴ S3(2)(a) of the Fertilizers, Farm Feeds, Agricultural Remedies and Stock Remedies Act, 1947

⁵ S3(4)(b) of the Fertilizers, Farm Feeds, Agricultural Remedies and Stock Remedies Act, 1947

- 14.1 Section 3 provides the information to be supplied by the applicant, including scientific documentation to demonstrate the safety, quality and efficacy of the agricultural remedy.
- 14.2 Sections 5 and 6 allows for the expedition of a review of an application for minor use or of low-risk agricultural remedy registration.
- 14.3 Section 10 provides that a certificate issued under section 3 will remain valid for a period of 3 years and may be renewed for an additional 3-year period.
15. Part III of the Regulations addresses the labelling requirements, with a general requirement that classification and labelling be in accordance with the Globally Harmonised System of Classification and Labelling of Chemicals (GHS)⁶ and the registration holder is responsible for ensuring the accuracy of the GHS classification of the agricultural remedy.
16. Part IV of the Regulations deals with the requirements that must be met for agricultural remedies to be imported into South Africa.
17. Part VII of the Regulation prohibits anyone from distributing or selling an agricultural remedy that has not been approved by the Register.

III International comparison of biopesticide regulatory frameworks

18. Below we compare seven regulatory frameworks being the United Kingdom (UK), European Union (EU), United States of America (USA), Canada, Australia and Brazil to South Africa. The purpose of the comparison is to ascertain best practices useful to informing a tailored biopesticide registration system ensuring alignment with global standards.
19. The legislative frameworks of Australia, the USA and Canada recognise biopesticides as a separate category to conventional chemical pesticides. The

⁶ ““Globally Harmonized System” “or GHS” means the Globally Harmonized System of classification and labelling of chemicals, a guidance document developed by the United Nations for standardising and harmonising the classification and labelling of chemicals globally, as may be updated from time to time, commonly known as the UN Purple Book;”

EU, UK and Brazil treat biopesticide within existing plant protection law. South Africa classifies biopesticides under the broad and vague term “Agricultural Remedy”; this has the effect of limiting promotion of novel and alternative biopesticide products.

20. In general, the regulation responsibilities fall to environmental or agricultural government departments. The USA (through its Environmental Protection Agency, Biopesticides and Pollution Prevention Division (BPPD)) and Canada (through its Pest Management Regulatory Agency) have dedicated units responsible for the review and approval of biopesticide dossiers. South Africa’s regulatory agency, Department of Agriculture lacks a dedicated and specific framework and departments or divisions for the registration of biopesticides contributing to an uncertain and lengthy registration process which ensures none or a very small number of biopesticides are registered.
21. The registration process within the USA, Canada and Australia have specific data requirements for the registration of biopesticides. Australia recognises that some data requirements might not be appropriate for specific products and urges applicants to apply for waivers, where suitable. Brazil has specific biopesticides that are exempt from registration all together and other products that are only subject to a simplified registration. The EU has a two-tier registration process and the UK’s registration process is similarly aligned with the EU. South Africa currently has no enacted law that establishes a separate and dedicated registration framework for biopesticides, registration of biopesticides is governed by the Regulations.
22. In terms of the registration period; the USA registration of biopesticides typically takes between 7 to 19 months, in the EU 4 to 5 years and Brazil 2 to 4 years. South Africa lacks a designated and streamlined process for the registration of biopesticides and so it is unclear what the registration period is for biopesticides.
23. In terms of protection goals; six of the countries have developed legislation that aims to ensure product safety for humans, non-target animals and the

environment with the EU and Australia having the most stringent registration criteria. South Africa requires the Registrar to be satisfied that the remedy is acceptable in the context of public interest⁷, suitability and efficacy. The Act itself at no point references health, safety or the environment other than the GHS labelling requirements.

24. The USA's EPA⁸ has specified measures that promote the use of biopesticides, including regulatory relief activities, shorter review times and less data requirements. Canada's PMRA⁹ encourages the development and implementation of innovative and sustainable pest management through implementation of policies. Brazil has passed law that encourages on farm production of bio-inputs and promotion of the adoption of science and innovation¹⁰. EU places restrictions on synthetic chemical pesticides¹¹ and require a drastic reduction in the use of chemical pesticides through its Farm to Fork strategy¹². South Africa currently offers no incentives or developmental support that foster the use of biopesticides in comparison.

25. Brazil's innovation and leadership for bioinput production is individually noteworthy, through its enactment of the Law No. 15070: Brazil's New Regulatory Framework for Sustainable Bioinputs ("New Bioinputs Law"). Brazil's regulatory model now creates a clearer regulatory environment, recognises the need for a specific and unique registration process, promotes the adoption of science and innovation and aims to provide farmers with viable and safe alternatives to conventional chemical pesticides.

⁷ In the absence of a definition in the Act itself, we borrow from the ideas of the Information Regulator who has recently submitted a basic formulation of the idea of public interest in a guidance note *"Public interest is a wide and diverse concept that cannot, and should not, be limited in its scope and application. The definition of what constitutes public interest varies across jurisdictions and should be assessed on a case-by case basis. In its very basic formulation public interest is the notion that an action or process or outcome widely and generally benefits the public at large (as opposed to a few or a single entity or person) and should be accepted or pursued in the spirit of equality and justice."*

⁸ Environmental Protection Agency

⁹ Pest Management Regulatory Agency

¹⁰ New Law 15070 enacted in Dec 2024

¹¹ Directive 2009/128/EC

¹² Farm to Fork Strategy implemented by the EU and aims to ensure food systems fair, healthy and environmentally friendly - [Farm to Fork Strategy - Food Safety - European Commission](#) accessed on 29 October 2025

26. The New Bioinputs Law defines bio-inputs as products, processes or technologies of plant, animal or microbial origin, including those from biotechnological processes, or structurally similar and functionally identical to those of natural origin, intended for use in the production, protection, storage and processing of agricultural products or in aquatic production systems or planted forests, which interfere with the growth, development and response mechanism of animals, plants, microorganisms, soil and derived substances and which interact with products and physical, chemical and biological processes. The New Bioinputs Law encourages the production of bioinputs for agriculture through:

- 26.1 Registration deadlines for low-risk bioinputs that decrease the estimated registration period;
- 26.2 Permits on-farm production of bioinputs exclusively for farmers personal use prohibiting commercialization;
- 26.3 Allows for exemption from registration for bioinputs that are used exclusively for personal use, certain semiochemicals and established low-risk products;
- 26.4 Provides a simplified administrative process for the registration of already registered comparable products;
- 26.5 A fee structure that supports the financing of inspection and control activities.
- 26.6 Other incentives introduced to encourage research, development, production and use of bioinputs include tax and credit, interest rates and public policies where priority is given to microenterprises, cooperatives, and family farmers.

27. Common challenges faced by South Africa include lack of capacity, training, lengthy and complicated approval processes, vague definitions, and little incentive.

IV Best practices for registration of biopesticides through comparison of foreign legislation:

28. Below we discuss the suggested best practices and frameworks that relate to the registration of biopesticides in the context of certainty, efficiency, effectiveness and advancement of the use of biopesticides.
29. The European Parliament's Committee on the Environment, Climate and Food Security report on "Ensuring the Faster Registration and uptake of Biological Control Agents" (the EU Report) recommends the following measures for the establishment of a separate legal category for biopesticides products:
- 29.1 Establishing a clear definition of biopesticides as well as a framework for the accelerated approval of biopesticide products. This framework must also contain clear data requirements and updated guidelines. This may be done through the promulgation of regulations and policies regarding the registration of these biopesticide products.
 - 29.2 The legal framework ought to understand that biopesticides have fundamentally different properties, risk profiles and modes of action to synthetic pesticides and therefore must be assessed accordingly.
 - 29.3 The effective evaluation of the stability of current risk assessments during the approval of biopesticides and their authorisation and if need be, develop a unique assessment protocol for biopesticide products.
 - 29.4 Support small to medium sized businesses in their application process through providing explanations of data requirements, submission procedures. This will increase the quality of applications and thus reduce backlogs and queries.
30. The US EPA categorizes biopesticides as pesticides derived from natural materials such as plants, animals, bacteria and minerals. These biopesticide products are defined clearly and undergo different registration processes to synthetic pesticides. This clear definition and separation of biopesticide products in the US allows for clarity in the registration, risk analysis and data pathways.¹³

¹³ <https://www.epa.gov/pesticide-registration/about-pesticide-registration>

31. However, the Federal Pesticide Laws and Associated statutes regulate the pesticides and enables the EPA to oversee the registration of pesticides. The Federal Insecticide, Fungicide and Rodenticide Act and the Federal Food, Drug and Cosmetic Act are the primary sources of law to enforce the rigorous regulation of pesticides.¹⁴

Dedicated pathway for registration of biopesticides

32. There needs to be a dedicated pathway for registration of biopesticides through simplified, expedited registration processes specifically for low-risk biopesticides.

33. The EU Report recommends the following fast-tracking procedures:

33.1 The streamlining, optimising and accelerating the authorisation and renewal procedure for biopesticides will reduce unnecessary administrative burdens, ensure legal evaluation deadlines are achieved and any associated risks are properly assessed.

33.2 These procedures should, however, not come at the expense of rigorous scientific assessment and analysis of risk.

33.3 Allow for the provisional authorisation of a biopesticides that fulfil the relevant safety criteria as per the report, and where conditional authorisation has been granted, necessary monitoring criteria should be included.¹⁵

34. The US EPA uses a separate set of requirements and guidance documents for the registration of biopesticides. This allows for the expedited registration of low-risk biopesticides. This is done through the publishing of clear eligibility criteria and timelines.¹⁶

35. The Australian government uses a similar model to the US in their registration of biopesticides through the Australian Pesticides and Veterinary Medicines

¹⁴ <https://www.epa.gov/pesticide-registration/about-pesticide-registration#compliance>

¹⁵

https://www.europarl.europa.eu/doceo/document/A-10-2025-0234_EN.html#:~:text=whereas%20there%20is%20often%20a,leadership%20in%20sustainable%20plant%20protection;

¹⁶ <https://www.epa.gov/pesticide-registration/biopesticides-confidential-statement-formula>

Authority (APVMA). This authority contains specified guidelines and delineates requirements for biopesticides to ensure smooth registration and use.¹⁷ The APVMA also uses a flexible approach to examine each biopesticide on a case-by-case basis, this limits rubber stamping and ensures that the APMA applies its mind to each biopesticide product registration.¹⁸

Policy integrity: biopesticides reform must not become a regulatory back door

36. A core objective of this submission is to enable South Africa to fast-track **only** low-risk, **naturally derived** biopesticides to reduce dependency on high-risk chemical pesticides **excluding genetically modified or gene-edited organisms and traits**, which must be dealt with under dedicated biosafety governance.

Building institutional capacity

37. There needs to be a dedicated panel within the regulatory agency that is responsible for building institutional capacity and training evaluators in biopesticide registration methods and risk analysis.

38. The EU report recommends the following for the building of capacity and registration methods of biological control products (referred to by this document and SADC as biopesticides):

38.1 Ensure that all administrative procedures are clear and time-efficient with specific help desks to ensure that applications are completed within a predetermined timeframe category dependent.

38.2 Administrators of the authorisation of biopesticides must undergo a performance evaluation and institutional capacities monitoring and evaluation must occur to ensure that all authorities are competent to fulfil their tasks.

¹⁷

https://www.apvma.gov.au/registrations-and-permits/data-requirements/agricultural-data-guidelines/biological?utm_source=chatgpt.com

¹⁸

https://www.apvma.gov.au/registrations-and-permits/data-requirements/agricultural-data-guidelines/biological?utm_source=chatgpt.com

- 38.3 The Treasury must allocate the necessary funding for training and employment of additional staff with the necessary skills.
- 38.4 The State should foster investment in public and private research and partnerships for the development of biopesticide products and uses.
39. The US EPA, Canada PMRA and the Australian APVMA all utilize ring fenced funding to build institutional capacity. This ensures that monies cannot be squandered through corruption, fruitless and wasteful expenditure or be used for other purposes.

Reducing Dependency on Chemical Pesticides

40. Efforts to reduce dependency on chemical pesticides through incentivised development, can include offering reduced registration fees, faster timelines for applications relating to biopesticides products, enabling private-public research cooperation and piloting field trials and supporting on-farm production, allowing small-scale or community led production with appropriate oversight.
41. The EU report recommends the following with regard to reducing dependency on chemical pesticides:
- 41.1 Training farmers to be able to use biopesticides, particularly regarding the care of beneficial species that protect crops and the approaches to the use of biopesticide products.
- 41.2 Allocating funds by treasury to accelerate the uptake of biopesticides. Treasury must ensure sufficient budget for the regulation authority to adequately process authorisation applications.

SADC Framework

42. The Harmonised Guidelines for the Registration of Biopesticide Products and Biological Control Agents in Southern Africa ("SADC Guidelines") provide a clear framework for universal implementation of harmonised guidelines throughout SADC member states, they emphasise the importance of managing and reducing negative health effects, whether acute or chronic, that

are faced by human beings, animals and the environment due to the use of conventional pesticides, they promote safer and efficacious pest control tools and methods that are low in toxicity and more sustainable and not only manage pests and disease but improve agricultural production, productivity and trade.

43. The SADC Guidelines recommend a framework that aims to expedite registration of biopesticides, facilitate mutual recognition between SADC member states for biopesticide registration and promote best agricultural and regulatory management practices for agricultural pest control. The framework includes:

43.1 definitions,

43.2 categories of biopesticides (microbial biopesticides, macrobials, semichemicals and botanicals. ***UnPoison vehemently excludes (PIPS) Plant Incorporated Protectants***),

43.3 registration categories (experimental use, temporary or emergency use, full registration and transit),

43.4 specific registration requirements for each category of biopesticides and

43.5 renewal and amendment processes.

South Africa needs to align with global standards.

44. In order to ensure alignment with global standards, it is imperative that these recommendations form the basis of legislation.

45. Additionally, collaboration between AU member states in their domestication of these frameworks can ensure regional accountability and harmonisation across borders through the standardisation of data requirements, evaluation formats and terminology. The adoption of the Organisation for Economic Co-Operation and Development (OECD) guiding documents and principles will form the basis for harmonisation if all are domesticated.

46. South Africa recognises biopesticides broadly through the inclusion of biological remedy in the definition of an agricultural remedy. There is no clear and separate legal distinction for biopesticides under the Act.
47. There are no tailored data requirements and legal processes that are clearly and separately set out for applicants to register biopesticides. Applicants are expected to follow the regulatory model for the registration of agricultural remedies which requirements are sometimes inappropriate or irrelevant.
48. There is currently a lack of harmonization with international standards for biopesticides.

V Key elements and recommendations

49. The Act requires reform and should incorporate appropriate language in all applicable sections of the Act¹⁹ ensuring a standalone legal category.
50. Commitment to the following protection goals:
- 50.1 sound management of biopesticides; and
 - 50.2 protection of human, animal and plant health, and the environment.
51. Informed by the SADC Guidelines, the below recommendations in respect of definitions are to be used as guidance:
- 51.1 **Biopesticide as** “A generic term generally applied to a substance derived from nature, such as a microorganism or botanical that may be formulated and applied in a manner similar to a conventional chemical pesticide and that is normally used for short-term pest control.”
 - 51.2 **Biological control as** “A pest control strategy that uses living natural enemies, predators, antagonists or competitors of target pests.”
 - 51.3 **Biological control agent as** “A natural enemy, predator, antagonist or competitor, and other self-replicating biotic entity used for pest management.”

¹⁹For example: s3 of the Act should be amended to include registration of biopesticide product(s) or biological control agents, s4 of the Act should be amended to include cancellation of licenses for biopesticide product(s) or biological control agents, s13 and s18 should be amended to incorporate offences and penalties applicable to biopesticide product(s) or biological control agents.

52. Informed by the SADC Guidelines, the following categories of biopesticides should be considered:

52.1 Microbial biopesticides:

Microbial biopesticides (non-GM): Products containing a, **or a combination of**, microorganisms (bacterium, fungi, virus, or protozoa) that produce a pesticidal effect that often include competition or inhibition, toxicity and even use of the target pest as a growth substrate. Microbial biopesticides can control many different kinds of pests, although each separate active agent may be relatively specific for its target pest(s).

52.2 Macroorganisms (Predators and Parasitoids):

Macroorganisms to be defined as: “Organisms large enough to be seen by the normal unaided human eye”;

Predator to be defined as: “A natural enemy that preys and feeds on other animal organisms, more than one of which are killed during its lifetime.”; and Parasitoid to be defined as: “An insect parasitic only in its immature stages, killing its host in the process of its development, and free living as an adult.

52.3 Biochemicals / Botanicals:

Botanicals to be defined as: “Natural (unmodified) plant extracts and formulations of pesticides that originate from plants as phytochemicals and elicit toxic effects on insects and mites.

Biochemical / Semio chemicals:

Biochemical / Semio chemicals to be defined as: “Products based on natural substances that have been manufactured by chemical synthesis processes and are intended to control insect target pests by altering the behavioral or physiological conduct of insect pests of same species or other species

53. Exclusion:

53.1 Products that are Plant-Incorporated Protectants (“PIPs”), contain genetically modified organisms, or rely on genetically engineered traits intended for pest control do not qualify for classification or registration under the biological inputs pathway. Such products must be regulated

under the Genetically Modified Organisms Act and any applicable biosafety requirements.

- 53.2 **The categories in this submission should be read as limited to naturally derived biological control approaches** (e.g., non-genetically microbial, macrobials; biochemical/ botanicals; biochemical/ semiochemicals), and should exclude PIPs and Genetically Modified Organisms (“GMOs”) by explicit drafting.

54. Exclusion of PIPs and GMO’s from the biological category.

- 54.1 **UnPoison submits that PIPs and any GMO’s (genetically modified microbes, genetically modified fungi/viruses, or other living modified organisms) must be explicitly excluded from the proposed biopesticide/biological category and any expedited registration pathway intended for naturally derived biopesticides.**

- 54.2 Distinct legal and biosafety governance is required for PIPs and GMO’s:

- 54.3 International biosafety law recognises that living modified organisms (LMOs/GMOs) present a distinct risk-management problem that requires case-by-case, scientifically sound risk assessment in the likely receiving environment, with specific methodological steps and points to consider. This is the core purpose of the Cartagena Protocol on Biosafety²⁰, which requires decisions on environmental introduction of LMOs to be based on such risk assessments.

55. Accordingly, a biopesticide pathway designed to accelerate low-risk, naturally derived inputs is not the appropriate legal vehicle for GM organisms or GM-derived pest control traits. Existing South African law already provides the correct framework:

- 55.1 South Africa’s Genetically Modified Organisms Act 15 of 1997 establishes measures and procedures specifically intended to evaluate and reduce risks arising from activities involving GMOs, including risk assessment criteria and institutional oversight mechanisms.

²⁰ Cartagena Protocol on Biosafety to the Convention on Biological Diversity, Convention on Biological Diversity, 2003

55.2 It is therefore legally and administratively inappropriate to “place” CRISPR and GMOs into a biological inputs track that is premised on fundamentally different assumptions about origin, persistence, and risk.

55.3 Scientific guidance distinguishes genetically modified microbial agents from conventional microbial biopesticides.

55.4 International regulatory science guidance recognises that genetically modified microbial biopesticides raise additional biosafety considerations (e.g., stability, persistence, ecological interactions, potential gene transfer concerns) that are not resolved by the assessment logic for naturally occurring microbial biopesticides. Organisation for Economic Cooperation and Development (“OECD”) guidance is designed for microbial biopesticide assessment generally, but regulatory practice clearly distinguishes where different (and more stringent) frameworks apply for genetically modified organisms.

55.5 Further, European Food and Safety Authority (“EFSA”) guidance on the risk assessment of genetically modified microorganisms and their food and feed products²¹ expressly notes that where a microorganism submitted for authorisation as a plant protection product is genetically modified, the assessment engages the interplay between pesticide regulation and GMO biosafety law.

56. Policy integrity: biopesticide reform must not become a regulatory back door

56.1 A core objective of these submissions is to enable South Africa to fast-track low-risk, naturally derived biopesticides to reduce dependency on high-risk chemical pesticides. Allowing PIPs or GMOs into the same category risks creating a **regulatory back door** whereby technologies requiring heightened biosafety scrutiny are processed through a framework designed to reduce burden and accelerate uptake. This would

²¹ Guidance on the risk assessment of genetically modified microorganisms and their food and feed products, European Food Safety Authority, 2011

undermine regulatory credibility, public confidence, and legal defensibility of the biopesticides reforms.

56.2 The Act envisions the appointment of a registrar who is responsible for registration, cancellation or discontinuation of agricultural remedies. The Minister is also empowered to designate persons including technical advisors and analysts of agricultural remedies²² It is submitted that through this framework, the Minister appoints a registration committee that facilitates the registration process, management and monitoring of biopesticides for the manufacturing, acquisition, disposal, sale or use within South Africa.

57. Specific regulations should be promulgated with the aim to:

- 57.1 provide an alternative to conventional pesticides which are safe to humans, animals and the environment;
- 57.2 provide a framework that is clear, certain and in harmony with
- 57.3 international standards;
- 57.4 facilitate best practice during registration including where possible
- 57.5 mutual recognition of product registration in certain countries;
- 57.6 where possible **provide incentive development.**

58. The scope of the regulation must apply to the registration process and post-registration process relating to manufacturing, importing, exporting, packaging, repackaging, storing, distributing, advertising and offering to sell or selling biopesticides within South Africa.

59. In line with the SADC guidelines, it is submitted that the general administrative information requirements should consist of:

- 59.1 Proof of payment of the appropriate application fee.
- 59.2 A cover letter outlining the purpose of the application.
- 59.3 A biopesticides registration application form.
- 59.4 Name, type of business, telephone, and physical address of the applicant or registrant.
- 59.5 Registration number of the applicant or registrant.
- 59.6 Local agent if different from applicant or registrant.

²² s14 of the of the Fertilizers, Farm Feeds, Agricultural Remedies and Stock Remedies Act, 1947

- 59.7 Product brand name.
- 59.8 Type of biopesticides.
- 59.9 New biopesticide or amended registration.
- 59.10 Product manufactured domestically or imported.
- 59.11 Intended uses.
- 59.12 Target pest(s) and crop or other use(s).
- 59.13 Hazard classification for the active ingredient according to WHO and GHS guidelines.
- 59.14 GHS classification for the formulated product.
- 59.15 Signed declaration certifying that the information presented, as well as the relevant technical data submitted in the dossier, in support of this application are true, correct and complete.

60. In line with the registration of an agricultural remedy the regulations should provide for:

- A. Specific set of data requirements, scientific evaluations and decision-making criteria relating to the aforementioned categories of biopesticides, which can be informed by the SADC Guidelines specifically articles 8, 9, 10, 11 and 12;
 - B. A general prohibition on the manufacturing, importing, exporting, packaging, repackaging, storing, distributing, advertising and offering to sell or selling biopesticides that have not been approved by the registration committee;
- 60.1 General labelling requirements and specific guidelines if necessary;
 - 60.2 Prioritisation of the registration of biopesticides;
 - 60.3 Commitment towards acceptance of scientific data / registrations from appropriate countries such as OECD members, in line with the Regulations, and SADC members, wherein which the committee may consider issuing requirement waivers / full approvals;
 - 60.4 Risk based assessments, efficacy requirements and requests for additional information;
 - 60.5 Where necessary, registration or use conditions;

60.6 Additional protection of confidential information if the application contains information that is considered confidential.²³

61. Informed by the ICGB review, the regulations should include various registration categories and data requirements necessary for each category.

62. Informed by the SADC and EAC Guidelines registration categories can consist of:

62.1 Experimental use permit (*“status granted (bearing experimental Permit number) allowing experimental use prior to submitting application for full registration”*);

62.2 Full registration (*“Granted when all registration requirements have been met and the registration authority has granted approval for biopesticides according to the conditions laid down in the certificate of registration”*);

62.3 In line with the Regulations:

62.3.1 Temporary or emergency registration (*“means a registration of an agricultural remedy that is granted under special circumstances for limited and controlled use, where such a registration is necessary because of urgency a danger which cannot be contained by any other reasonable use²⁴”*). It is submitted that emergency registration be added to the list of registration categories provided that such use will not cause unreasonable negative effects to human or animal health or the environment.

62.3.2 in-transit registration- to be granted in advance of arrival for the unloading, landing or other movement of biopesticides through a SADC Member State and not intended for domestic distribution and sale.”

63. An approval process should be detailed including:

²³ section 17 of the Act: *“Any person who discloses except to the Minister or to any other person for the purpose of the performance of his duties or the exercise of his functions under this Act or when required to do so by any court or under any law, any information acquired by him in the exercise of any duty or function under this Act, in relation to the business or affairs of any other person shall be guilty of an offence and liable on conviction to a fine not exceeding five hundred pounds or imprisonment for a period not exceeding twelve months.”*

²⁴ s1 of the Regulations relating to Agricultural Remedies, 2024

- 63.1 issuing of committee's decision including approval, rejection or imposition of conditions
- 63.2 issuing of a certificate of registration;
- 63.3 prescribed application fees for each category;
- 63.4 prescribed timelines for application decisions to be observed by the committee;
- 63.5 duration of validity of the registration, it is suggested 3 years for a full registration in line with the Regulations²⁵;
- 63.6 publication of the approved license, subject to applicable confidentiality requirements.

64. A renewal process should be detailed including:

- 64.1 effect of expiration of a registration. It is submitted that registrations which have not been renewed should be deemed to have been cancelled;
- 64.2 prescribed application fees and requirements for renewals, including but not limited to the original / certified copy of the certification of registration;
- 64.3 prescribed timelines for renewal decisions to be observed by the committee;
- 64.4 declaration confirming:
 - 64.4.1 no new evidence has come to light which may impact the result of the risk assessment specifically towards the health and safety of humans, animals and the environment;
 - 64.4.2 no deviation whatsoever from the original details, in relation to the issuing of the committee's decision including approval, rejection or imposition of conditions;
- 64.5 publication of the approved renewed license, subject to applicable confidentiality requirements.

65. An amendment process to the certificate of registration:

- 65.1 distinction between minor²⁶ or major amendments

²⁵ s10(1) of Regulations relating to Agricultural Remedies, 2024

²⁶ section 1 of the Regulations relating to Agricultural Remedies, 2024

65.1.1 minor changes are changes that have no change to their composition such as registration holder's details, adding or changing emergency numbers, changes in brand names, artwork changes.

65.1.2 major changes all other changes for example change to composition.

65.2 submission requirements for minor²⁷ and major amendments prescribed fees, approval, decision-making process, timelines to be observed by the committee and publication of the approved amendments, subject to applicable confidentiality requirements.

66. License to manufacture, import or export:

66.1 specific prohibition against importation / exportation / manufacture of biopesticides unless such registration has been approved and the applicant is in receipt of the appropriate certificate.

66.2 specific requirements for the application such as:

66.2.1 prescribed fees;

66.2.2 prescribed form;

66.2.3 necessary details for example²⁸:

66.2.3.1 details of the applicant, manufacturer,

66.2.3.2 purpose of importation,

66.2.3.3 storage facilities,

66.2.3.4 proof of foreign registration certificate,

66.2.3.5 proof of compliance with appropriate legislation and regulations, for example labelling and packaging requirements

66.2.3.6 copy of safety data sheet;

66.2.3.7 publication of license grant;

66.2.3.8 if designed for exportation, registration status of destination country.

²⁷ section 8(4): "If the holder of an agricultural remedy registration notifies the Registrar of an administrative minor change and provides a declaration confirming that no other changes in the registration details have been made, the Registrar shall advise the registration holder approval for such change has been granted."

²⁸ Details for importing and manufacturing can be informed by part IV and V of the of the Regulations relating to Agricultural Remedies, 2024

- 66.2.4 application conditions for importation, if appropriate;
- 66.2.5 importation may only be from the manufacturer set out in the application;
- 66.2.6 validity period of license;
- 66.2.7 compliance requirements to international registration requirements, if appropriate;
- 66.2.8 obligation to ensure and maintain appropriate staff, premises, equipment and facilities to ensure functions are carried out correctly and in line with guidelines;
- 66.2.9 **harmonization and data sharing should be a factor to consider for example granting of such licenses in line with any relevant international / regional agreements.**
- 66.2.10 approval process and specific ports of entry²⁹
- 66.2.11 publication of the approved license to manufacture, import or export, subject to applicable confidentiality requirements.

67. Informed by part IX and XI of the Regulations, necessary records and returns to be furnished for example:

- 67.1 section 36 “Audited sale records”
- 67.2 section 37 and 38 “Reporting on adverse reactions”. Adverse reactions include any adverse effect or risk of the biopesticide product to human health or the environment or where the biopesticide product no longer complies with the certificate of registration.
- 67.3 sections 42 and 43 “Appeal against decisions of the committee”

68. Obligation to facilitate cooperation and capacity / resource mobilisation to ensure proper management of biopesticides including surveillance, monitoring, data collection, compliance and data sharing, where appropriate.

²⁹ section 28 “(1) Unless the Registrar directs otherwise, no person shall import any agricultural remedy in terms of section 16 of the Act into the Republic of South Africa except through one of the following ports of entry—

- (a) Cape Town International Airport or Cape Town harbour;
- (b) Port Elizabeth International Airport or Port Elizabeth harbour or COEGA harbour;
- (c) King Shaka International Airport or Durban harbour;
- (d) Richards Bay harbour; and
- (e) O.R. Tambo International Airport.”

69. Other consideration to be included:

69.1 Suspension procedures in certain instances for example:

- 69.1.1 upon request by the applicant / license holder;
- 69.1.2 the information supplied by the applicant is misleading / inconsistent with product;
- 69.1.3 the product has been mislabelled / altered without appropriate approvals;
- 69.1.4 unexpected negative effects to human or animal health or the environment;
- 69.1.5 publication of the cancelled license, subject to applicable confidentiality requirements;
- 69.1.6 non-compliance with conditions set out in the license;
- 69.1.7 in the event of importation, use of a manufacturer not specified in the license.
- 69.1.8 The effect of a suspended license should be detailed.

69.2 Disposal obligations, which can be informed by requirements set

- 69.2.1 out in part VII of the Regulations³⁰, also including:
- 69.2.2 obligation to dispose of product is placed on the holder of the license; and
- 69.2.3 disposal to occur with an inspector present.

69.3 Incentive and/or post-registration programmes consideration of incentives such as:

- 69.3.1 reduced registration fees;
- 69.3.2 exemption processes where appropriate for example with low-risk products;
- 69.3.3 increased validity periods for licenses.
- 69.3.4 post-registration programmes should be developed:
 - 69.3.4.1 to enable public-private cooperation and research for example through pilot trials;
 - 69.3.4.2 promote development and integration into IPM plans;³¹

³⁰s34 Regulations relating to Agricultural Remedies, 2024

³¹ Integrated Pest Management

69.3.4.3 to create an institutional framework for accreditation and certification³².

69.4 Advertising requirements including but not limited to the requirement to comply with FAO guidance and GHS labelling guidance, which can be informed by the Regulations.

69.5 Transparency requirements through publication of a public registry and non-confidential summaries with guidance documentation and checklists.

69.6 Offences and penalties.

Recommendations are in line with the following foreign guidelines:

70. In 2023 SADC developed a draft of harmonised guidelines which aimed to harmonise the registration of Biopesticide Products and Biological Control Agents in Southern Africa, ("**SADC guidelines**") in order to provide farmers with effective biopesticides that are safe for the environment and public health, sets out a framework for SADC regions to use in order to fast-track biopesticide registration, facilitate mutual recognition of data between SADC member states and promote best practices for biopesticides (Link: [Harmonized Guidelines for the Registration of Biopesticide Products and Biological Control Agents in Southern Africa](#)).

71. In 2014 the Association of Southeast Asian Nations (ASEAN) submitted guidelines for the regulation, use and trade of biological control agents ("**ASEAN Guidelines**"). These guidelines aim to provide a general universal framework to better implement biopesticides, harmonise regulations, provide essential data requirements for registration and post-registration compliance obligations and highlight the importance of oversight by regulatory authorities with regards to product scrutinization of labels. (Link: [Implementing Biological Control Agents in the ASEAN Region](#)).

72. In 2019, the East African Community submitted guidelines for the registration of biopesticides, underpinned by the need to ensure safer pesticide products

³² National Programme for Organic Production (NPOP) (2001)

within the region (“**EAC Guidelines**”) the EAC Guidelines aim to ensure access to safer and effective biopesticides. (Link: [BIOPESTICIDES AND biopesticide AGENTS GUIDELINES](#)).

73. In 2022, the International Centre for Genetic Engineering and Biotechnology (“**ICGEB**”) published a review of the biopesticide regulatory frameworks in Botswana, Mozambique, South Africa, Tanzania, Zambia and Zimbabwe, created under the framework: “*Enhancing trade through regulatory harmonisation and biopesticide-based residue mitigation in the SADC region*”, the paper reviewed key factors to be implemented in order to achieve regulatory harmonisation of biopesticides (“**ICGEB review**”).

VI Conclusion

South Africa’s current regulatory framework for biopesticides is inadequate. In order to align with global standards and promote safer alternatives, urgent legislative reform is required. This includes establishing a distinct legal category for biopesticides, creating streamlined registration processes and building institutional capacity. Harmonization with international guidelines and incentivizing biopesticide adoption will not only protect human, animal and environmental health but also foster innovation and competitiveness in the agricultural sector.

Implementing these measures will ensure an environmentally responsible regulatory system that meets constitutional obligations and advances food security. A proposed operational architecture to give effect to these reforms is set out in Annexure 1 - Drafting Annex For Biopesticide Registration Pathway.



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Annexure 1: Drafting Annex For Biopesticide Registration Pathway

1. Scope and Definitions Hierarchy

1.1. Scope statement for drafting

This framework applies to biopesticides (biological plant protection products) used to prevent, repel, attract, suppress, or control pests, weeds, or plant diseases in agriculture, public health environments, and related settings.

This framework does not regulate: (i) biofertilisers and biostimulants (unless pesticidal claims are made), which should follow a distinct biological inputs pathway; and (ii) Plant-Incorporated Protectants (PIPs) and genetically modified organisms (GMOs) / engineered pesticidal traits, which must be regulated through South Africa's GMO/biosafety framework and must not be processed through the expedited biopesticide pathway.

1.2. Definitions hierarchy

- 1.2.1. Biopesticide: A generic term generally applied to a substance derived from nature, such as a microorganism or botanical that may be formulated and applied in a manner similar to a conventional chemical pesticide and that is normally used for short-term pest control
- 1.2.2. Active ingredient: the biologically active component responsible for pesticidal function (including microorganism, semiochemical, botanical extract, metabolite).
- 1.2.3. Formulated product: Any biopesticide product containing one or more active ingredients mixed with other components in a form in which it is packaged and sold and can then be used to be applied to a specific target

1.2.4. Biopesticide category groupings :

- 1.2.4.1. Microbial biopesticides (non-GM): Products containing a, **or a combination of**, microorganisms (bacterium, fungi, virus, or protozoa) that produce a pesticidal effect that often include competition or inhibition, toxicity and even use of the target pest as a growth substrate. Microbial biopesticides can control many different kinds of pests, although each separate active agent may be relatively specific for its target pest(s).
- 1.2.4.2. Biochemical/Botanicals: Natural (unmodified) plant extracts and formulations of pesticides that originate from

plants as phytochemicals and elicit toxic effects on insects and mites.

- 1.2.4.3. Biochemical / Semio chemicals to be defined as:
“Products based on natural substances that have been manufactured by chemical synthesis processes and are intended to control insect target pests by altering the behavioral or physiological conduct of insect pests of same species or other species
- 1.2.4.4. Macroorganisms: An invertebrate natural enemy used for pest management.

Drafting note: Where a product could straddle categories (e.g., microbial metabolite), classification must follow the primary mode of action and exposure profile.

2. Registration Workflow and Decision Points

2.1. Pre-submission support

Step 0: Pre-submission meeting (voluntary). Applicants may request a meeting to confirm classification, tier, and data requirements. The Department of Agriculture, Land Reform and Rural Development (DALRRD) issues a written “data expectations note” within 10 (ten) working days after the meeting.

2.2. Application intake and screening

Step 1: Submission. Applicant submits dossier electronically (and hard copy where required), including classification claim (biopesticide category + tier request as described below), product identity and composition, proposed label and claims, and supporting studies/literature.

Step 2: Administrative screening (completeness check). Registrar screens for completeness within 10 (ten) working days of receipt of the dossier.

Outcomes: accepted for evaluation with or without conditions, or returned as incomplete with a checklist of deficiencies.

2.3. Scientific evaluation and risk-tier confirmation

Step 3: Classification + Tier confirmation. DALRRD confirms biopesticide category and risk tier and may reassign tier where exposure/uncertainty is higher than claimed.

Step 4: Scientific evaluation (risk-proportionate). Evaluation covers identity & quality, hazards, exposure & risk characterisation under realistic conditions, efficacy, environmental fate/ecotoxicity (as relevant), and label adequacy and risk mitigation.

2.4. Public participation trigger (only where required)

Step 5: Public comment (Tier 2–3 only, as described below; or where novel/high-interest). Public notice plus access to a non-confidential summary. 30 days for comment (extendable by Registrar for good cause). Registrar retains discretion over late comments.

2.5. Decision and publication

Step 6: Decision. Registrar issues approval (with or without conditions), conditional approval (label changes/additional monitoring/restricted uses), or refusal with written reasons and appeal pathway.

Step 7: Registry entry. All approvals entered into a public registry (minimum: product name, active, category, permitted uses, approval date, expiry date, key restrictions).

3. Risk Tiers and Timeframes

- 3.1. Tier 1: Low-risk / expedited (target 90 working days from acceptance of a complete dossier).
- 3.2. Tier 2: Moderate-risk / standard biological evaluation (target 180 working days).
- 3.3. Tier 3: Higher-uncertainty / full evaluation (target 365 working days).
- 3.4. Stop-the-clock rule: DALRRD may stop timelines while awaiting applicant responses, with written notice.

4. Data Requirements and Data Waiver Logic

4.1. Core dossier (all categories, all tiers)

- 4.1.1. Identity and composition (active + co-formulants);
- 4.1.2. Manufacturing process summary and quality controls;
- 4.1.3. Batch consistency / certificate of analysis parameters;
- 4.1.4. Proposed label (claims, directions, PPE, disposal, storage);
- 4.1.5. Efficacy evidence proportionate to claims and context of use;
- 4.1.6. Safety summary and risk mitigation measures;
- 4.1.7. Non-confidential summary for public registry

4.2. Category add-ons (minimum expectations)

- 4.2.1. Microbial biopesticides (non-GM): identity/strain, contaminants, viability/shelf-life, metabolite profile where relevant, operator exposure and sensitisation considerations, non-target assessments proportional to use pattern.
- 4.2.2. Biochemical/Botanicals: compositional fingerprint, source/extraction method, impurities profile, stability/storage, residues considerations where relevant, non-target/ecotoxicity proportional to exposure.
- 4.2.3. Biochemical/Semiochemicals: purity and release technology, species specificity and non-target behavioural impacts, negligible toxicity rationale (often waiver-eligible), fate proportional to use.
- 4.2.4. Macroorganisms: species ID/origin, rearing QA, host range/specificity, ecological establishment risk, permit/containment conditions where relevant, post-release monitoring where warranted.

4.3. Data waiver principles

DALRRD shall waive or reduce data requirements where an applicant demonstrates history of safe use, negligible exposure, well-characterised identity, and/or international approvals with equivalent data standards (subject to local relevance). DALRRD shall not waive core identity/quality data or any data necessary to resolve plausible non-target harm, persistence/bioaccumulation, sensitisation/allergenicity (where relevant), or food residue risks for edible crops.

5. Post-Registration Controls

Approvals should be valid for 3 years, with expedited renewal where formulation/claims are unchanged and no adverse signals exist. Registrants must report serious adverse events in humans/animals, significant non-target impacts, contamination or batch failures, and loss-of-efficacy patterns consistent with resistance (where relevant) within defined timeframes. DALRRD retains powers to suspend/withdraw approvals and enforce compliance.

6. Institutional Arrangement

A dedicated Biologicals Evaluation Unit (or panel) should include, as needed by category: microbiology/virology expertise; ecotoxicology/environmental fate; occupational exposure/health; agronomy/plant pathology; and regulatory decision

writing capacity. Transparency should be ensured through publication of a public registry and non-confidential summaries, with guidance documents and checklists.

7. Exclusion Rule (PIPs and GM agents)

Plant-Incorporated Protectants (PIPs) and genetically modified organisms (including genetically engineered pesticidal traits expressed in organisms intended for release/use) are excluded from this biopesticide pathway and shall be governed by the GMO/biosafety framework, with dedicated risk assessment and permitting requirements.